The Algorithmic Big Bang: Cosmogenesis, Inflation, and the Arrow of Time at the Univalent Horizon

Halvor S. Lande
hsl@awc.no

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Abstract

General Relativity fails at $t = 0$, yielding nonsensical infinite values at the initial singularity. Physics currently lacks a mathematical vocabulary for a universe assembling its own rules. We demonstrate that the initial singularity is not a point of infinite mass-energy density in pre-existing spacetime, but the exact computational phase transition where external metamathematical assembly (algorithmic time, τ) is compiled into internal geometric dynamics (physical time, t). By applying the Kinematic-Dynamic Equivalence ($\bigcirc X \simeq X^{\mathbb{D}}$) of the Dynamical Cohesive Topos at Step 15 of the PEN Generative Sequence [1], we show that the Big Bang represents the ignition of autonomous guarded corecursion. This algorithmic origin naturally derives the thermodynamic arrow of time, the minimal-entropy initial state, and provides a strict type-theoretic mechanism for cosmic inflation. The Univalent Horizon—the structural boundary at which the sequence permanently halts—is identified as the cosmological firewall separating the pre-physical compilation epoch from the self-computing physical universe.

1 Introduction: Cosmogenesis as a Logical Phase Transition

1.1 The First-Cause Paradox and Its Dissolution

Standard cosmological models treat time as a pre-existing real line $\mathbb{R}$ upon which a singularity is placed at $t = 0$. The singularity is then a pathology: a point where curvature scalars diverge, geodesics are incomplete, and the field equations cease to predict [3]. This framing converts an explanatory failure into an ontological posit—the universe "began" in a state that the theory's own mathematics cannot describe.

The difficulty is structural, not technical. Any framework that presupposes a continuous time manifold and then asks what happened at its boundary is asking the container to explain its own origin. The question "what caused the Big Bang?" inherits the full weight of the first-cause paradox: every causal explanation requires a prior temporal context, generating an infinite regress or an unexplained brute fact.

PEN dissolves this paradox by rejecting the premise. Time is not an ambient container in which events are arranged; it is a data structure that had to be algorithmically constructed and selected for efficiency [1]. The Generative Sequence builds spatial topology, cohesion, differential structure, metrics, and functional-analytic machinery across 14 discrete steps *before* temporal dynamics exists. Physical time—continuous, geometric, directional—arrives at Step 15 as the output of the Kinematic-Dynamic Equivalence ($\bigcirc X \simeq X^{\mathbb{D}}$), not as a background assumption.

1.2 Compiler vs. Runtime

The conceptual structure of PEN's cosmogenesis maps precisely onto the distinction between compilation and execution in computation theory.

Definition 1.1 (Compilation Epoch)**.** The *compilation epoch* (Steps 1–14) is the discrete meta-mathematical assembly of the kinematic framework. During this epoch, the library $\mathcal{B}_n$ grows monotonically under the selection dynamics (Axioms 1–5 of [1]), accumulating type formers, higher-inductive types, modal operators, and axiomatic extensions. No continuous internal observers, trajectories, or dynamical equations exist. Time is measured solely by algorithmic ticks τ_n and integration latency $\Delta_n = F_n$.

Definition 1.2 (Runtime Ignition)**.** *Runtime ignition* (Step 15) is the selection of the temporal-cohesive shell whose semantic completion is the Dynamical Cohesive Topos (DCT): it introduces temporal modalities ($\bigcirc$, $\Diamond$), explicit exchange laws with cohesion, and guarded coherence clauses into the library. At this step, the universe transitions from extrinsic structural selection (maximizing ρ) to intrinsic dynamical evolution (integrating vector fields on $\mathcal{U}$).

The Big Bang, on this account, is neither a physical explosion nor a mathematical singularity. It is the exact moment at which a self-assembling formal system becomes causally closed and begins computing its own future.

2 The Pre-Physical Epoch: Algorithmic Time (τ)

2.1 Assembly of the Phase Space

Prior to Step 15, the library $\mathcal{B}_{14}$ contains the full spatial and functional-analytic infrastructure of modern physics: dependent types, higher-inductive types (S^1, S^2, S^3), cohesion ($\flat, \sharp, \mathrm{Disc}, \Pi_{\mathrm{coh}}$), connections, curvature, metrics, frame bundles, and Hilbert functionals [1]. These structures exist as a static, well-typed formal library—a univalent 0-groupoid sequence of realized specifications. They describe *what kinds of spaces and operations are available*, not *how anything moves within them.*

The distinction is categorical: $\mathcal{B}_{14}$ is a kinematic library, not a dynamical system. It contains the type S^3 but no trajectory on S^3; it contains the metric tensor $g_{\mu\nu}$ but no Einstein equation governing its evolution; it contains Hilbert spaces but no Schrödinger equation propagating states through them. The library is a phase space without a Hamiltonian—a stage fully constructed but with no actors yet in motion.

2.2 The Two Clocks

A persistent conceptual error in cosmological thinking is the implicit assumption that "before the Big Bang" must refer to a prior interval of physical time. PEN makes the temporal structure fully explicit by distinguishing two incommensurable notions of time.

Definition 2.1 (Algorithmic Time)**.** *Algorithmic time* τ is the discrete tick counter of the Generative Sequence. At each tick τ_n, the selection dynamics evaluates candidates against the bar $\mathrm{Bar}_n = \Phi_n \cdot \Omega_{n-1}$, realizes the minimal-overshoot winner, and increments the library. The elapsed algorithmic time at step n is $\tau_n = \sum_{i=1}^{n} \Delta_i$, where $\Delta_i = F_i$ is the Fibonacci integration latency.

Definition 2.2 (Physical Time)**.** *Physical time* t is the continuous geometric parameter arising from the Kinematic-Dynamic Equivalence (Lemma 5.10 of [1]):

$$\bigcirc X \;\simeq\; X^{\mathbb{D}}. \tag{1}$$

Physical time exists only after Step 15, when the "next" modality $\bigcirc$ is identified with the tangent endofunctor $(-)^{\mathbb{D}}$, converting discrete temporal succession into continuous infinitesimal displacement.

The two clocks have entirely different character. Algorithmic time is discrete, external to the library, and measured in structural ticks. Physical time is continuous, internal to the DCT, and measured in infinitesimal tangent displacements. There is no meaningful sense in which τ_{14} is "earlier than" $t = 0$; the two quantities inhabit different ontological categories, related only by the phase transition at Step 15 that converts one into the other.

Remark 2.3 (Why "before the Big Bang" is a category error). Asking "what happened before the Big Bang" presupposes that physical time extends to the left of $t = 0$. In PEN, the pre-Step 15 epoch is not a prior interval of t but a structurally distinct generative process indexed by τ. The question dissolves: nothing happened *before* $t = 0$ in physical time, because physical time does not exist before Step 15. What happened *instead* of physical time was algorithmic assembly—a categorically different process that produced the conditions under which physical time could exist at all.

3 Step 15 Ignition: The Birth of Physical Time ($t = 0$)

3.1 The Arrival of LTL

The Step 15 temporal-cohesive shell extends the library $\mathcal{B}_{14}$ with four structural components [1]:

(1) **Spatial Logic:** The cohesive interface already present in $\mathcal{B}_{14}$, namely ($\flat \dashv \sharp$, $\Pi_{\text{coh}} \dashv \text{Disc}$).
(2) **Temporal Core:** The modalities $\bigcirc$ ("next") and $\Diamond$ ("eventually") from Linear Temporal Logic, following Nakano [5].
(3) **Spatial-Temporal Exchange:** Explicit commuting laws between temporal and cohesive structure, exemplified by $\bigcirc(\flat X) \simeq \flat(\bigcirc X)$ and $\sharp(\Diamond X) \simeq \Diamond(\sharp X)$.
(4) **Guarded Coherence:** The temporal self-interaction and elimination clauses that make the shell executable as a typed AST.

The specification cost is $\kappa = 8$, audited as the eight temporal-cohesive clauses of the selected AST [1]. In the current strict machine canon, the generative capacity is $\nu = 88$, yielding $\rho = 11.00$ and clearing the Step 15 bar of 7.51 by the largest absolute overshoot in the sequence [1]. The infinitesimal object $\mathbb{D}$ belongs to the semantic completion of this shell rather than to the primitive Step 15 syntax.

3.2 The Kinematic-Dynamic Equivalence as Geometrization of Time

The central mechanism is the Kinematic-Dynamic Equivalence: in the semantic completion of the temporal-cohesive shell, the discrete temporal modality $\bigcirc$ is naturally equivalent to the continuous tangent endofunctor $(-)^{\mathbb{D}}$ [1].

This is not a primitive stipulation of the selected AST but a *forced semantic identification.* The exchange laws with the cohesive modalities and the guarded coherence clauses constrain $\bigcirc$ so strongly that, in the DCT reading, it is represented by tangent exponentiation. The proof does not begin by postulating $\mathbb{D}$ in the syntax; rather, $\mathbb{D}$ is derived as the infinitesimal shadow of the first-order temporal-cohesive endofunctor selected by the engine [1].

The physical content of this equivalence is profound: *the passage of time is the same operation as infinitesimal spatial displacement.* What the temporal logic calls "the next state" is what synthetic differential geometry calls "an infinitesimal tangent vector." This identification establishes $t = 0$—not as a point on a pre-existing timeline, but as the moment at which the concept of temporal succession acquires geometric content.

3.3 The Guarded Corecursion Engine

The temporal logic of Nakano [5] carries a powerful recursion principle: Löb's guarded fixpoint rule,

$$\mathrm{fix} : (\bigcirc A \to A) \to A. \tag{2}$$

This rule states: given a function that can produce an A provided it has access to "tomorrow's" A, one can construct a definite A today. The guardedness condition (the $\bigcirc$ on the left of the arrow) prevents circular self-reference by ensuring that present computation may depend on future values only through the temporal delay.

Applied to the DCT's kinematic library, Löb's rule has the following operational effect. A discrete-time dynamical system is a map $f : X \to \bigcirc X$—given a current state, compute the next state. Under the Kinematic-Dynamic Equivalence, this becomes:

$$f : X \to X^{\mathbb{D}}, \tag{3}$$

which is precisely a *vector field* in synthetic differential geometry: a section of the tangent bundle assigning an infinitesimal displacement to each point. The guarded corecursion principle then unfolds any such vector field into a global trajectory:

$$\mathrm{unfold}_f : X \ \to \ \mathrm{Stream}\, X. \tag{4}$$

This is the cosmological spark. The 14 static structures of $\mathcal{B}_{14}$ are the fuel; $\bigcirc$ is the ignition mechanism; Löb's rule is the engine that converts a local infinitesimal prescription into perpetual, autonomous motion. Before Step 15, the library describes spaces. After Step 15, the library computes trajectories through those spaces.

Proposition 3.1 (Step 15 as the onset of dynamics)**.** *Prior to the realization of the DCT, no structure in $\mathcal{B}_{14}$ can express temporal evolution: there is no type former producing streams, no fixpoint principle, and no identification of succession with tangent displacement. The DCT is the* unique minimal *extension of $\mathcal{B}_{14}$ that introduces autonomous dynamics, and it does so with $\kappa = 8$ clauses.*

Proof. Dynamics requires (i) a notion of succession, (ii) a fixpoint or corecursion principle to unfold finite prescriptions into infinite trajectories, and (iii) compatibility with the existing geometric library so that trajectories inhabit the already-constructed spaces. Requirements (i) and (ii) are supplied by $\bigcirc$ and Löb's rule; requirement (iii) is enforced by the temporal-cohesive exchange laws and guarded coherence clauses. No proper subset of these eight clauses suffices: removing any temporal modality eliminates succession; removing the exchange laws severs temporal evolution from the geometric library; removing guarded coherence destroys the executable dynamics. The infinitesimal interpretation is then recovered semantically from this shell rather than paid for as a primitive clause. □

4 Cosmological Inflation as Combinatorial Schema Synthesis

4.1 The $\rho = 11.00$ Efficiency Peak

The strict machine-verified efficiency ratio at Step 15 is $\rho = 88/8 = 11.00$, clearing the bar (7.51) by an absolute overshoot of 3.49—the largest in the entire 15-step sequence [1]. This "efficiency singularity" is the PEN counterpart of the cosmological singularity, but with a crucial difference: the mathematics does not break down. Where General Relativity produces divergent curvature scalars at $t = 0$, PEN produces a finite, well-defined maximum of algorithmic efficiency. The singularity is not a failure of the formalism but a signature of *optimal synthesis*—the point at which the library's combinatorial capacity is most powerfully leveraged.

Remark 4.1 (Efficiency singularity vs. curvature singularity). In GR, the initial singularity signals the breakdown of the theory: physical quantities (density, temperature, curvature) diverge, and the Einstein equations lose predictive power [3, 4]. In PEN, the "singularity" at Step 15 is a computable finite maximum of a well-defined ratio. It arises not from a failure of description but from the temporal-cohesive shell's measured interaction payload with the existing library. The manuscript now treats the strict machine total $\nu = 88$ as canonical and no longer promotes the older structural-AST tally to theorem-level status [1]. The analogy is precise: both mark the moment of cosmological origin. The disanalogy is equally precise: one is a breakdown, the other is a peak of algorithmic performance.

4.2 The $\nu = 88$ Explosion and the Inflationary Epoch

Cosmic inflation is the hypothesis that the very early universe underwent a brief period of exponentially rapid expansion, during which the number of accessible physical degrees of freedom increased by many orders of magnitude [6, 7]. In PEN, the analogous phenomenon is the combinatorial schema explosion at Step 15.

The mechanism is the Combinatorial Schema Synthesis theorem [1]: when the DCT's temporal modalities ($\bigcirc$, $\Diamond$) are introduced into a library of 14 structures already equipped with spatial modalities ($\flat, \sharp, \mathrm{Disc}, \Pi_{\mathrm{coh}}$), the cross-interaction at depth 2 produces a superlinear expansion of new inference schemas. In the current strict lane, this temporal-cohesive shell is evaluated at $\kappa = 8$ and yields the canonical machine value $\nu = 88$. Representative schema families include:

- Guarded temporal endomorphism and recursion patterns built from $\bigcirc$ and $\Diamond$.
- Spatial-temporal exchange laws that tie the new temporal operators to the pre-existing cohesive modalities.
- Transport and flow-like interaction with the already selected cohesive and differential layers.

The picture is one of sudden, explosive logical cross-interaction: 14 structures that previously coexisted as independent library entries are now woven into a single dynamical fabric, with each pair generating new eliminators, new homotopy schemas, and new compositional types. This is the algorithmic analogue of inflationary reheating: the degrees of freedom of the physical universe are not injected from outside but are *released* by the synthesis of previously disconnected structure.

Proposition 4.2 (Structural inflation precedes thermal inflation)**.** *The combinatorial amplification at Step 15 ($\nu = 88$, compared to $\nu \leq 63$ at all prior steps) is a necessary precondition for any physical inflationary dynamics. Inflation requires a scalar field propagating on a differentiable manifold with Lorentzian metric; all of these structures are in $\mathcal{B}_{14}$, but they acquire dynamical content only at Step 15. The "inflaton field" is not a separate posit: it is the generic vector field $v : \mathcal{U} \to \mathcal{U}^{\mathbb{D}}$ whose existence is guaranteed by the DCT's corecursion principle.*

5 The Arrow of Time and Minimal Entropy

5.1 The Causal Arrow from Guardedness

Why does time flow in one direction? In thermodynamic accounts, the arrow of time is derived from the second law and the assumption of a low-entropy initial condition—but this merely pushes the question back to why the initial condition was special. In PEN, the arrow of time is not a thermodynamic consequence but a *type-theoretic primitive*: it is built into the logical structure of the $\bigcirc$ operator.

The guardedness condition in Löb's rule (2) is asymmetric by construction: a term of type $\bigcirc A \to A$ can use tomorrow's value to compute today's value, but there is no dual rule $A \to \bigcirc A$ that would allow today's value to determine tomorrow's independently of the dynamical law.

More precisely, the only way to produce a term of type $\bigcirc A$ is by applying a dynamical system $f : A \to \bigcirc A$ to a current state; one cannot extract a future state except by running the system forward.

Theorem 5.1 (Type-theoretic irreversibility)**.** *In the DCT, the guarded corecursion principle* $\mathrm{fix} : (\bigcirc A \to A) \to A$ *is not invertible: there is no general term of type* $A \to (\bigcirc A \to A)$ *that recovers the generating function from its fixpoint. Consequently, the temporal unfolding* $\mathrm{unfold}_f : X \to \mathrm{Stream}\, X$ *is causally one-directional: given a stream, one can extract any finite prefix by iterated application of* $\bigcirc$*, but one cannot in general recover the state at tick* n *from the state at tick* $n + k$ *for* $k > 0$*.*

Proof sketch. Suppose for contradiction that a general inverse $\mathrm{fix}^{-1} : A \to (\bigcirc A \to A)$ existed. Then for any type A, one could construct a term of type $\bigcirc A \to A$ from any element of A, which when fed back into fix would yield a fixpoint independent of the dynamical law f. This violates the productivity guarantee of guarded corecursion: the $\bigcirc$ guard ensures that each step of the unfolding makes genuine computational progress. A general inverse would collapse the guarded modality, making $\bigcirc A \simeq A$, which contradicts the non-triviality of temporal logic in the DCT (the temporal-cohesive exchange laws require $\bigcirc$ to remain distinguishable from the identity). □

Remark 5.2 (Closed timelike curves are ill-typed). A closed timelike curve (CTC) in General Relativity is a worldline that returns to its own past. In the DCT, a CTC would require a term of type $\mathrm{Stream}\, X \to X$ that extracts a state and feeds it back as the input to its own causal past—a circular dependency of the form $x = f(\bigcirc x)$ without the Löb guard. Such a term is *ill-typed* in the guarded fragment: the type-checker rejects it because the $\bigcirc$ delay cannot be discharged without forward progress. CTCs are not physically impossible for contingent dynamical reasons; they are logically excluded by the universe's native type system.

5.2 The Past Hypothesis as Algorithmic Compression

The "Past Hypothesis"—the empirical observation that the universe began in a state of extraordinarily low entropy [8, 9]—is one of the deepest unexplained facts in physics. Statistical mechanics can explain why entropy increases *given* a low-entropy initial state, but it cannot explain why the initial state was special. The low-entropy boundary condition is typically imposed by hand as a cosmological axiom.

PEN provides a constructive explanation. The $t = 0$ state—the state of the universe at the moment of runtime ignition—is not an arbitrary initial condition. It is the *output* of 15 steps of strict efficiency maximization under Kolmogorov-complexity-minimal specification.

Theorem 5.3 (Minimal algorithmic entropy at $t = 0$)**.** *The physical state at the onset of dynamics* ($t = 0$) *is specified by the complete DCT library* $\mathcal{B}_{15}$*, whose total specification complexity is* $\kappa_{\text{total}} = \sum_{i=1}^{15} \kappa_i = 92$ *clauses (equivalently, 610 bits in MBTT encoding [1]). This is the shortest description consistent with the full kinematic framework. Any reordering or alternative branch of the Generative Sequence that produces equivalent kinematic capacity requires equal or greater total* κ*.*

Proof sketch. The selection dynamics at each step chooses the candidate with minimal overshoot (Axiom 4 of [1]), which jointly minimizes κ subject to the bar-clearing constraint (ties broken by minimal κ). The result is a constructive proof that the 15-step trajectory is Kolmogorov-optimal among trajectories producing equivalent kinematic content: any alternative path that arrives at the same library must match or exceed the total specification cost, because each step is locally minimal in κ among bar-clearing candidates. □

The physical consequence is direct: the $t = 0$ state inherits the algorithmic compressibility of the Generative Sequence. A state that can be specified in 610 bits admits essentially no combinatorial disorder. It is a state of maximal algorithmic compression—which is the information-theoretic definition of minimal entropy.

Remark 5.4 (From algorithmic compression to thermodynamic entropy). The connection between Kolmogorov complexity and thermodynamic entropy is well-established in algorithmic information theory [10]: a microstate's thermodynamic entropy is bounded below by its algorithmic complexity. A system whose complete specification requires only 610 bits occupies a region of phase space so small that its Boltzmann entropy $S = k_B \ln W$ is negligible. The Past Hypothesis is therefore not a mysterious fine-tuning of initial conditions but the *inevitable* consequence of the universe's kinematic framework having been generated by an efficiency-maximizing compression process.

6 The Univalent Horizon as the Cosmological Firewall

6.1 Resolving the Halting Problem: Structural Termination

The PEN Generative Sequence does not "die" at Step 15. It halts—in the precise sense of Theorem 5.12 of [1]—because the external generator is rendered obsolete.

The argument has three steps:

1. The DCT internalizes its own evolutionary mechanism as a geometric flow on $\mathcal{U}$ (Lemma 5.11 of [1]). The metamathematical process of "searching for the next type" is representable as a vector field $v : \mathcal{U} \to \mathcal{U}^{\mathbb{D}}$, unfolded into a trajectory by guarded corecursion.
2. By univalence, paths in $\mathcal{U}$ are type equivalences: $(A =_{\mathcal{U}} B) \simeq (A \simeq B)$. Integrating the vector field produces a continuous family of types connected by paths in $\mathcal{U}$, hence a family of pairwise equivalent types. The flow is permanently confined to the connected component of $\mathcal{U}$ containing $\mathcal{B}_{15}$.
3. Any candidate Y proposed at Step 16 is either (a) internally derivable (contributing $\nu = 0$) or (b) structurally alien to the $d = 2$ Coherence Window (incurring $\Delta_{16} = F_{16} = 987$ integration latency, making $\rho \ll \mathrm{Bar}_{16} \approx 8.80$). In either case, the bar is permanently uncleared.

The universe does not stop evolving at Step 15; it stops *needing external structural input* at Step 15. The distinction is between a system that cannot continue (failure) and a system that has achieved self-sufficiency (closure). The Generative Sequence halts because the DCT is a fixpoint of the selection dynamics, not because the dynamics has run out of candidates.

6.2 Epistemic Isolation: The Cosmological Microwave Background Analogy

The Big Bang is opaque to physical observation: electromagnetic radiation from before the epoch of recombination ($t \approx 380{,}000$ years) is thermalized into the Cosmic Microwave Background [11], and no photon signal penetrates earlier. PEN provides a structural analogue of this observational horizon that is deeper than thermalization.

Proposition 6.1 (Epistemic isolation of the compilation epoch)**.** *Internal observers—agents whose perceptual and computational apparatus is itself a structure within $\mathcal{B}_{15}$—cannot construct an observable that probes the discrete algorithmic steps (1–14) that built the topos. The pre-Bang compilation epoch is accessible only via type-theoretic deduction, not physical measurement.*

Proof sketch. An "observation" in the DCT is a continuous geometric flow: a section of some bundle, a functional on a Hilbert space, or a path in a moduli space. All such flows are terms in the internal language of the DCT, and by the Univalent Horizon (Theorem 5.12 of [1]), they are confined to the connected component of $\mathcal{U}$ containing $\mathcal{B}_{15}$. The discrete algorithmic steps

$\tau_1, \ldots, \tau_{14}$ are *not* connected to $\mathcal{B}_{15}$ by paths in $\mathcal{U}$—they belong to the external metatheory, not the internal geometry. Any attempt to construct a physical observable that "sees" the compilation epoch would require a path crossing the Univalent Horizon, which is precisely what the horizon forbids. □

Remark 6.2 (The CMB as a physical shadow of the Univalent Horizon). The Cosmic Microwave Background is the earliest physical signal accessible to electromagnetic observation. In PEN, it is a *physical consequence* of the same structural boundary that the Univalent Horizon enforces type-theoretically. The opacity is not accidental—it is the observable signature of the fact that physical measurement (continuous geometric flow) cannot reach the discrete compilation epoch. The CMB is, in this sense, a thermal echo of the phase transition from algorithmic time τ to physical time t.

7 Conclusion

The standard approach to cosmogenesis looks for a physical, kinetic mechanism that *triggered* the Big Bang—a quantum fluctuation, a tunneling event, a collision of branes. Each such proposal presupposes the very structures (quantum fields, spacetime, dynamics) whose origin it is supposed to explain.

PEN inverts the explanatory order. The Big Bang is not an event that happened within physics; it is the mathematically necessary boundary at which physics *begins*. Specifically:

1. **The compilation epoch** (Steps 1–14) assembles the kinematic framework—types, spaces, cohesion, metrics, Hilbert functionals—in discrete algorithmic time τ, without presupposing continuous dynamics.
2. **Runtime ignition** (Step 15) introduces the DCT, whose Kinematic-Dynamic Equivalence ($\bigcirc X \simeq X^{\mathbb{D}}$) geometrizes time, converting discrete temporal logic into continuous infinitesimal structure. Löb's guarded corecursion sets the static library into perpetual, autonomous motion.
3. **The inflationary epoch** is the combinatorial schema explosion ($\nu = 88$) produced by the sudden cross-interaction of 14 previously disconnected kinematic structures.
4. **The arrow of time** is the type-theoretic asymmetry of the guarded $\bigcirc$ modality: forward computation is productive, backward extraction is ill-typed.
5. **The Past Hypothesis** is the algorithmic compressibility of the Generative Sequence: a universe built by 15 steps of Kolmogorov-optimal selection begins in a state of minimal entropy by construction.
6. **The Univalent Horizon** is the cosmological firewall separating the pre-physical compilation epoch from the self-computing physical runtime. The pre-Bang universe is epistemically isolated from internal observers, accessible only through the type-theoretic deduction that PEN provides.

The physical universe is the runtime execution of the Dynamical Cohesive Topos. We must stop looking for a first cause within physics and instead recognize the Big Bang as the halting condition of the universe's foundational compilation phase—the moment at which the formal system achieves self-sufficiency and begins computing its own future.

References

[1] H.S. Lande, "The Principle of Efficient Novelty: The Algorithmic Origin of the Kinematic Framework of Physics," `pen_unified.tex`, 2026.

[2] H.S. Lande, "Structural Emergence of 4D Lorentzian Geometry from the Principle of Efficient Novelty," companion paper, 2026.

[3] S.W. Hawking and G.F.R. Ellis, *The Large Scale Structure of Space-Time*, Cambridge University Press, 1973.

[4] R.M. Wald, *General Relativity*, University of Chicago Press, 1984.

[5] H. Nakano, "A Modality for Recursion," *Proceedings of the 15th Annual IEEE Symposium on Logic in Computer Science (LICS)*, 2000.

[6] A.H. Guth, "Inflationary Universe: A Possible Solution to the Horizon and Flatness Problems," *Physical Review D* 23 (1981), 347–356.

[7] A.D. Linde, "A New Inflationary Universe Scenario: A Possible Solution of the Horizon, Flatness, Homogeneity, Isotropy, and Primordial Monopole Problems," *Physics Letters B* 108 (1982), 389–393.

[8] R. Penrose, *The Road to Reality*, Jonathan Cape, 2004.

[9] S.M. Carroll, "The Cosmological Constant," *Living Reviews in Relativity* 4 (2001), 1.

[10] W.H. Zurek, "Algorithmic Randomness and Physical Entropy," *Physical Review A* 40 (1989), 4731–4751.

[11] A.A. Penzias and R.W. Wilson, "A Measurement of Excess Antenna Temperature at 4080 Mc/s," *Astrophysical Journal* 142 (1965), 419–421.

[12] The Univalent Foundations Program, "Homotopy Type Theory: Univalent Foundations of Mathematics," Institute for Advanced Study, 2013.